

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.45663690981713195, median 0.37838311349299875, std: 0.3460063278619112
Gyroscope error (imu0): mean 4.75296769734178, median 2.5375627135712397, std: 8.797480716629739
Accelerometer error (imu0): mean 10.819142775810628, median 4.910949526109016, std: 21.20909960872338

Residuals

Reprojection error (cam0) [px]: mean 0.45663690981713195, median 0.37838311349299875, std:
0.3460063278619112
Gyroscope error (imu0) [rad/s]: mean 0.0027265041983748806, median 0.001455653779439946, std:
0.005046608695116534
Accelerometer error (imu0) [m/s²]: mean 0.07665514065630745, median 0.03479476465747144, std:
0.1502694388445681

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[-0.02669641 -0.99938813 -0.02259806 0.03936745]
 [ 0.01563605 0.02218588 -0.99963158 0.06905726]
 [ 0.99952129 -0.02703991 0.0150342 -0.13412807]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam0 to imu0):

```
[[-0.02669641 0.01563605 0.99952129 0.13403504]
 [-0.99938813 0.02218588 -0.02703991 0.03418445]
 [-0.02259806 -0.99963158 0.0150342 0.07193796]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.07904849341877808

Gravity vector in target coords: [m/s²]

```
[ 0.13334062 1.61606961 -9.67155428]
```

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [158.2678720339621, 158.4685617430192]

Principal point: [158.94481680502764, 159.20472502261538]

Distortion model: radtan

Distortion coefficients: [-0.0009783416672607026, -0.00423095162741076, -0.0005988133129714586, -0.0022010262836639543]

Type: checkerboard

Rows

Count: 8

Distance: 0.02 [m]

Cols

Count: 11

Distance: 0.02 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

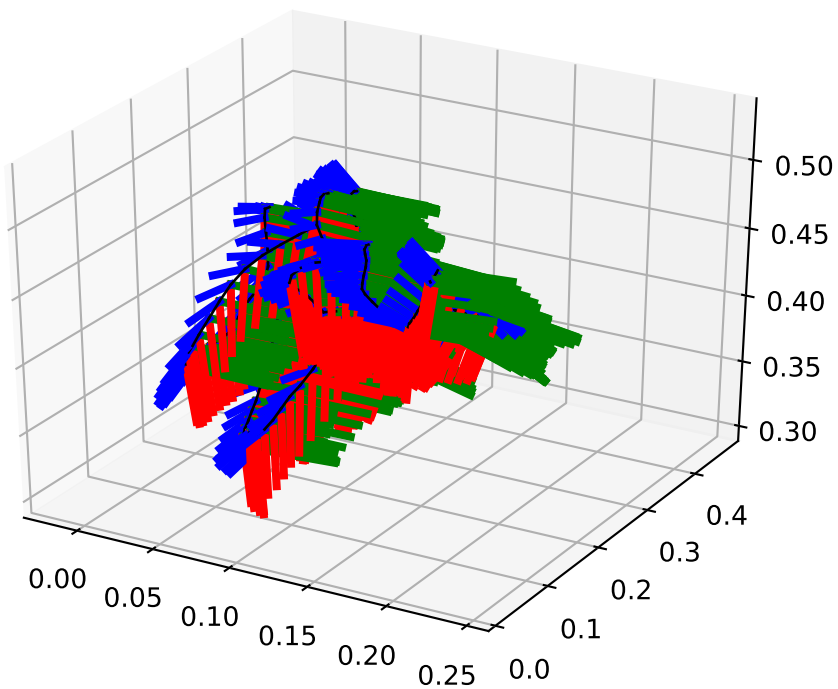
[0. 1. 0. 0.]

[0. 0. 1. 0.]

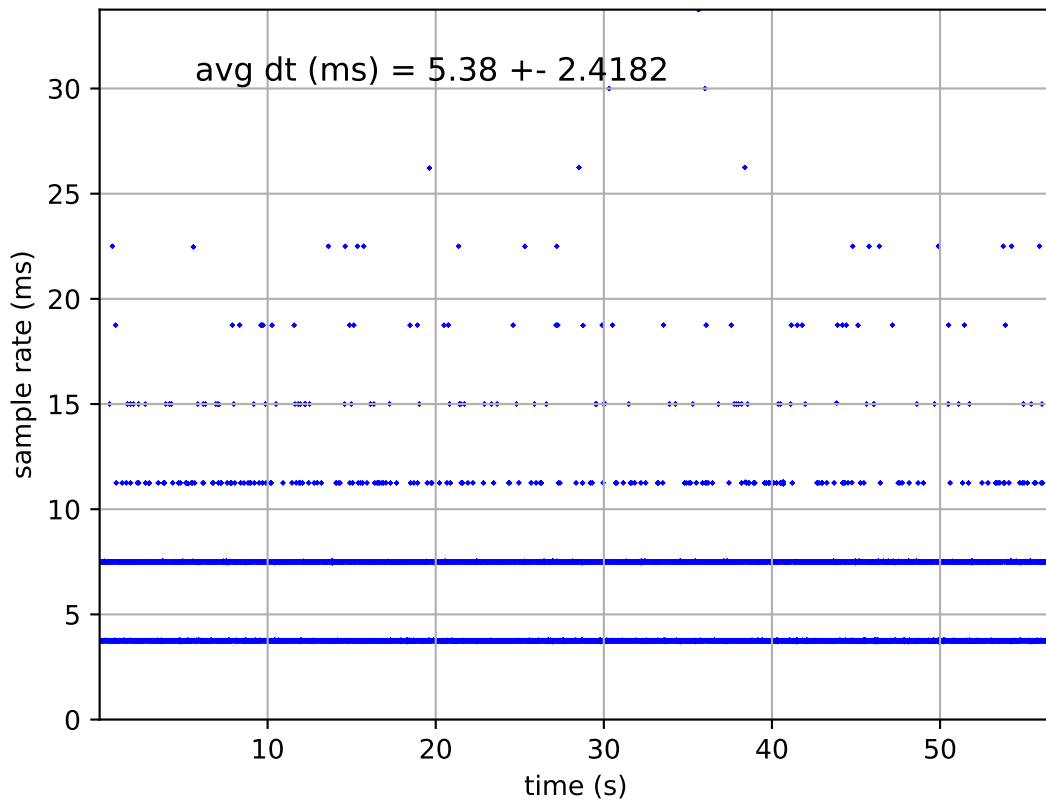
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

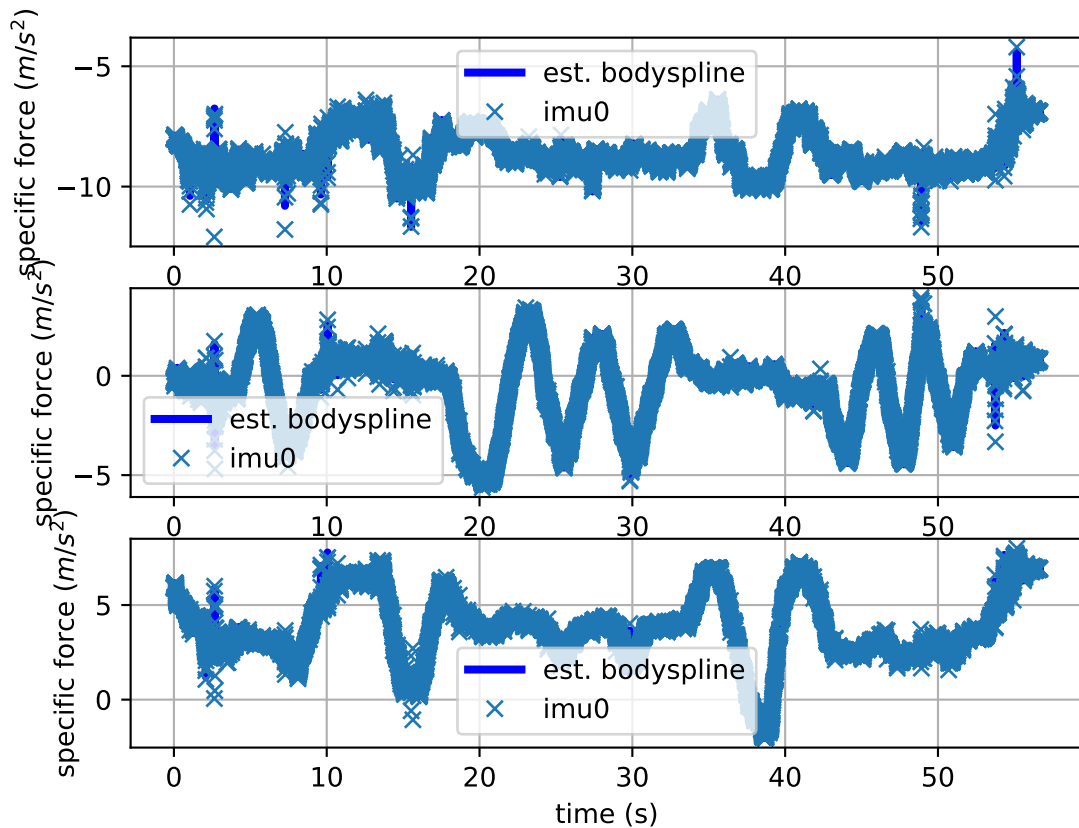
imu0: estimated poses



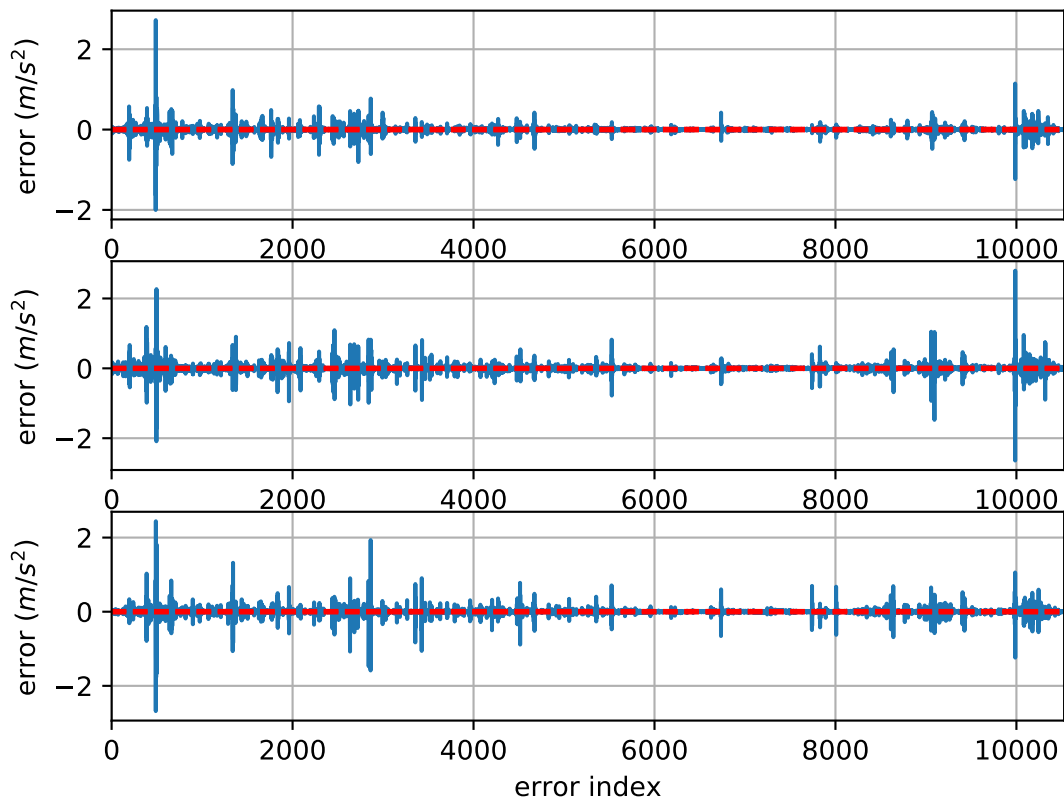
imu0: sample inertial rate



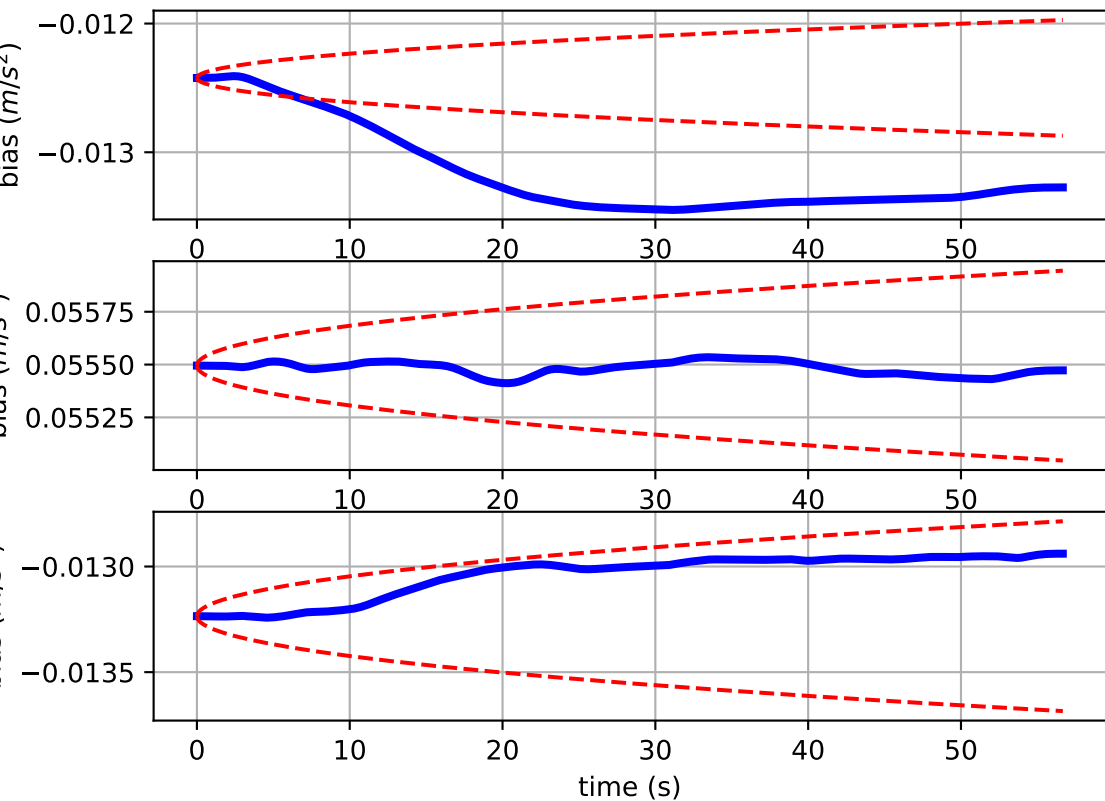
Comparison of predicted and measured specific force (imu0 frame)



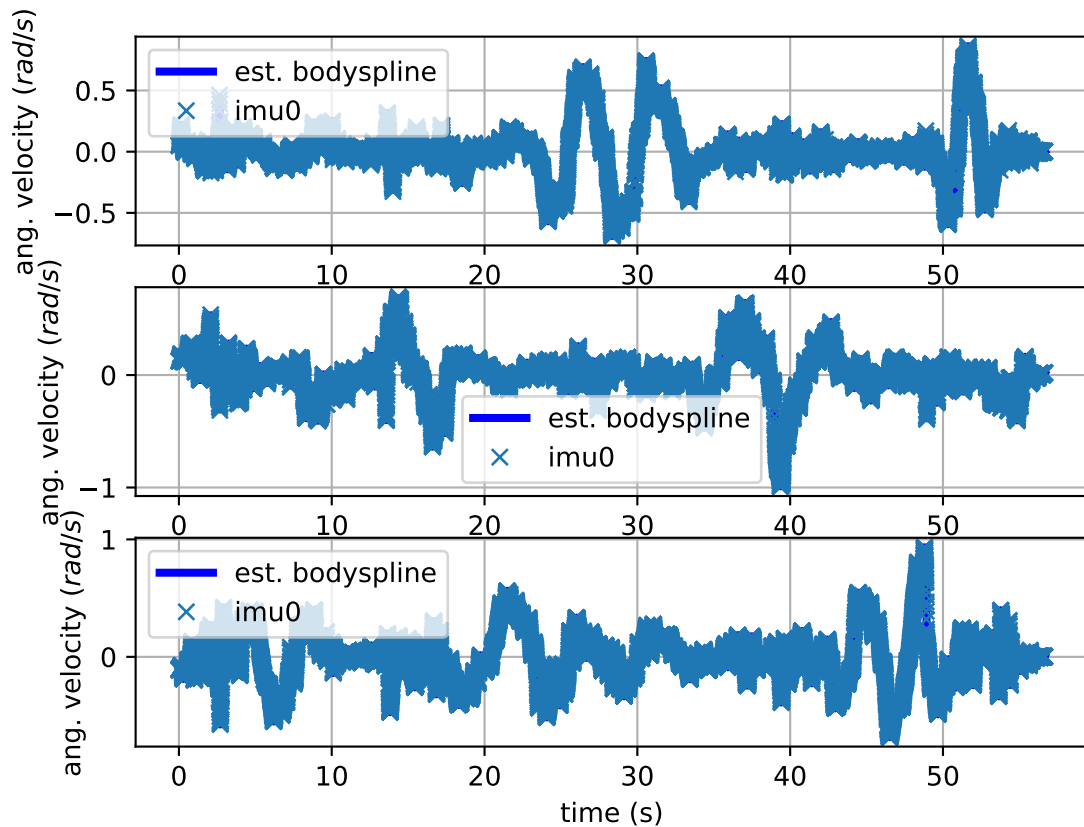
imu0: acceleration error



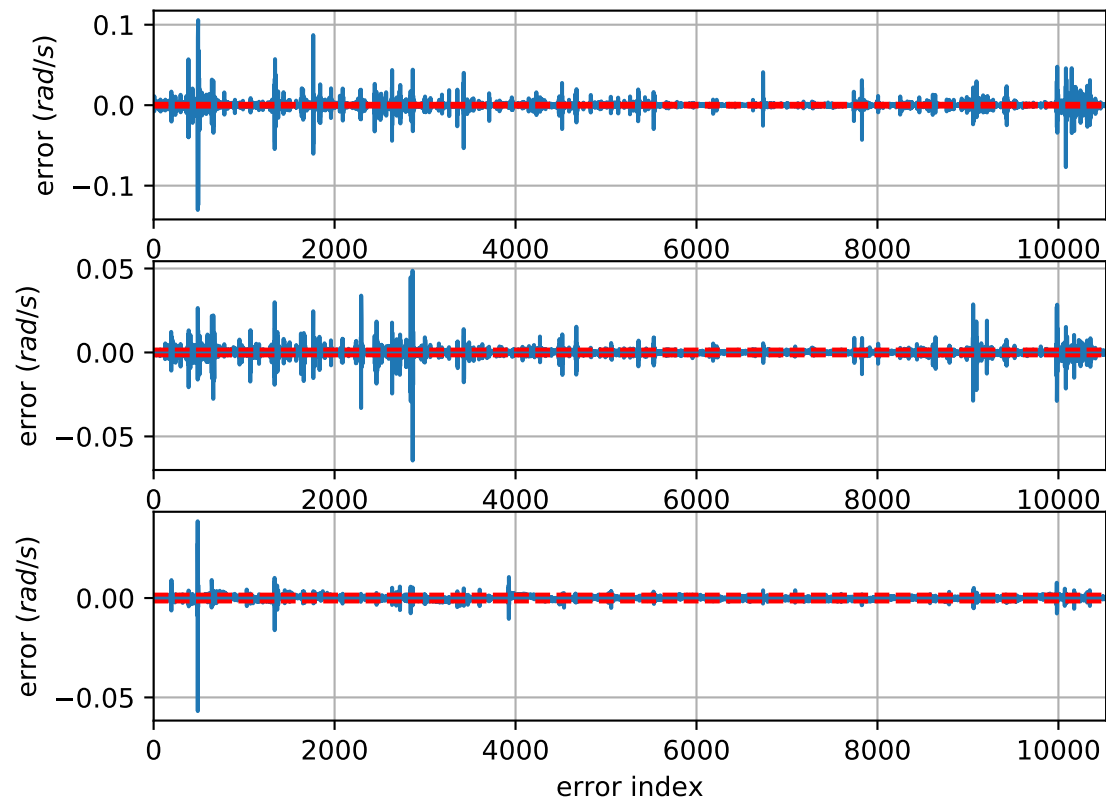
imu0: estimated accelerometer bias (imu frame)



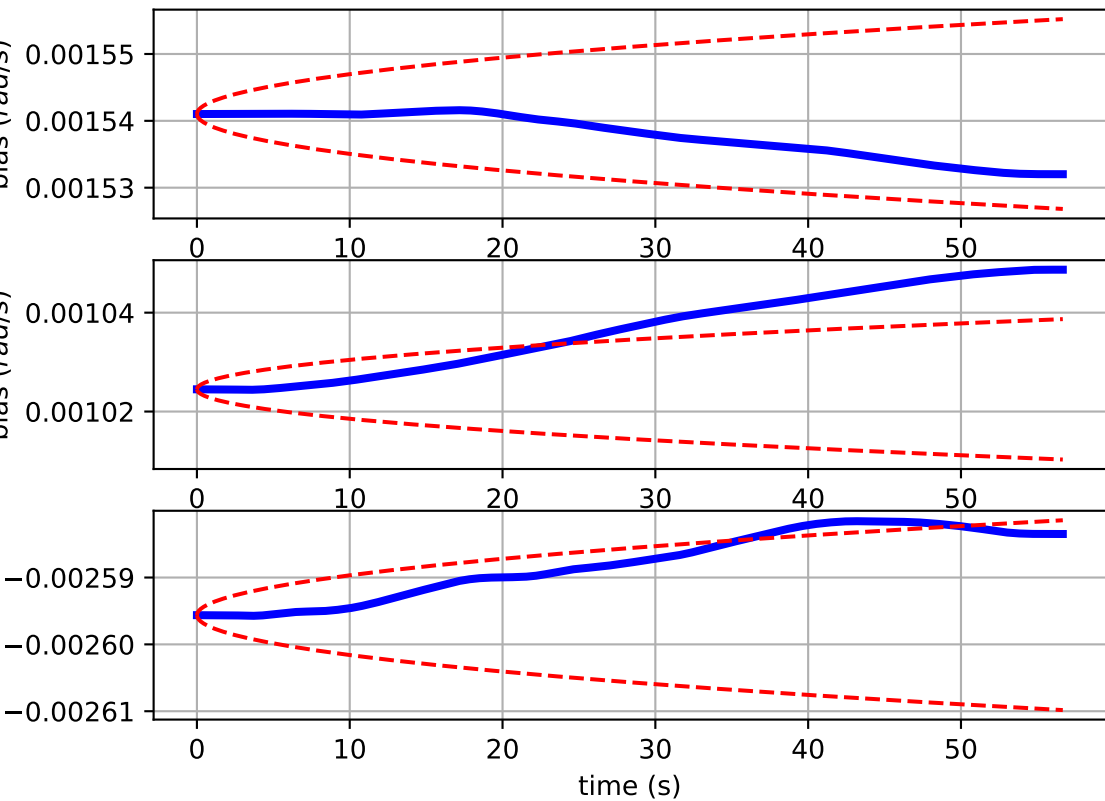
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

